



MINING DEALER BEST PRACTICE

# Cat 6015 - Disassembly of shovel final brake

**Maintenance and  
Repair**

Application

Component Life  
Management

Component  
Renewal (CRC)

MARC  
Management

|                                              |     |
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**February 2021**  
**1.04-210211-1**

## 1.0 Introduction

The Cat 6015 Hydraulic Front Shovel final drive brakes are an important part with a complex removal process.

The Caterpillar procedure (BI644878) recommends using two bolts with their M8 nuts by pressing the washer (thrust collar) while tightening the nuts uniformly to extract the retaining ring (Circlip).

The implementation of the bolts does not guarantee that the surface goes down uniformly every time which generates an abrasive wear between the piston and the housing (stripes). This could alter the reliability of the repair of the component and the reusability of parts if not done precisely. It could also increase the time under repair when the tightening is not adequate, or the applied loads are not uniform. This best practice can also reduce risk of the part ejecting out in an uncontrolled manner.

Parts reusability for damage to the piston at the submitting dealership is around 70%.



**Figure 1 - Bolts used in current procedure.**

Pull the piston (pos. 2.10) with 2 threaded rods and 2 nuts size M8 against the thrust collar (pos.1.200) by evenly tightening the nuts.

**Warning:** the pressure ring and piston are now spring-loaded.

**Figure 2 - Taken from the CAT procedure.**

This Best Practice aims to be able to disassemble the brake safely and reliably from a tool, preventing the washer ( thrust collar) from injuring the technicians, and it is also possible to eliminate the redo and cost overrun in case something comes up.

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## 2.0 Best Practice Description

This tool applies to the disassembly and assembly process on the final drive brakes on a 6015. It is geometrically "t-shaped". The vertical piece is a worm drive of 1" UNC that at one end has a hollow cylinder to rest on the pressure washer (thrust collar). The horizontal piece is a flat base of A36 steel that has a nut fixed in the center where the bolt travels; In addition, it has two holes of 22mm diameter at its ends designed to improve the grip between the tool and the final drive, ensuring the safety of the technicians and the component.



**Figure 3 - Tool for the brake disassembly and assembly process.**

## 3.0 Implementation Steps

1. Final Drive measurements are taken.
2. Design of the tool according to the measurements and ergonomics of the operation.
3. Load analysis of the tool.
4. Tool manufacturing.
5. Test conducted at the shop.
6. Start-up, after disclosing it to the operation with its change format.

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7. Operation of the tool (See Figure 4)



Figure 4 - Tool Operation

- 8. The tool is inserted in the final drive.
- 9. Place the bolts on the ends of the flat base in order to fasten them to the final drive in the holes marked in red (Figure 5).



Figure 5 - Bolt placement

- 10. Make sure that the tool is properly adjusted to the component.

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11. Start rotating the worm drive in order to lower the hollow cylinder of the tool against the Thrust Collar, compressing the piston springs, releasing the Circlip.



**Figure 6 - Tool Operation (continued)**

12. Remove the retainer ring.



**Figure 7 - Clip ready for removal**

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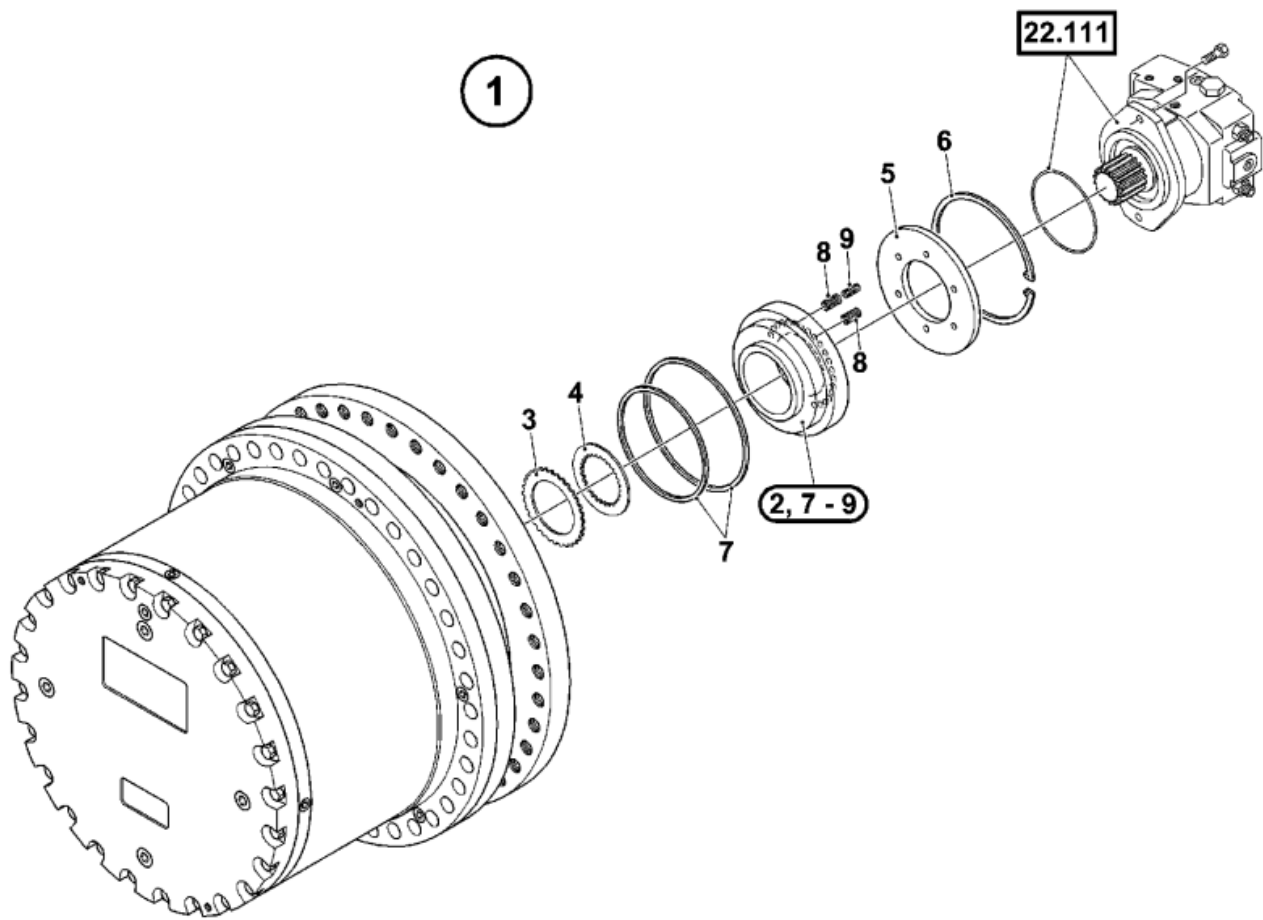


Figure 8 - Reference of Disassembly

4.0 Benefits

- Minimized risk of parts ejecting while disassembling.
- Reduction in 70% as of the disassembly and installation time of the brakes in the final drives.
- Elimination of redoes due to failures in the thread of the M8 bolts.

5.0 Resources Required

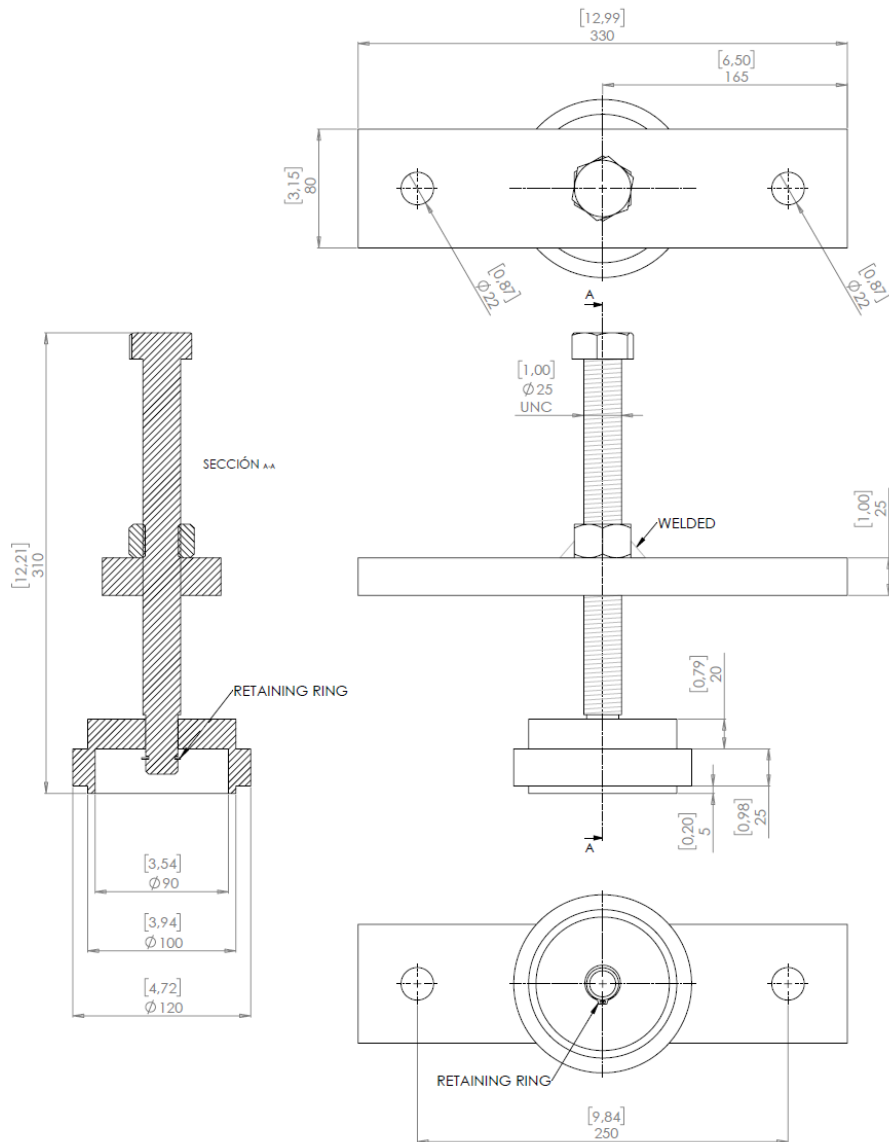
The investment regarding the manufacturing process of the tool was \$200 USD.

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## 6.0 Supporting Attachments / References

CAT procedure for assembly and disassembly of the 6015 Hydraulic Front Shovel final drives (BI644878).

DRAWING:



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00

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## 7.0 Related Best Practices

None

## 8.0 Acknowledgements

Enrique Urbina  
CRC Development Project Chief RELIZAN CAT  
Email: [enrique.urbina@relianz.com.co](mailto:enrique.urbina@relianz.com.co)

Juan Sebastián De la Hoz  
CRC Service Engineer RELIZAN CAT  
Email: [juan.delahoz@relianz.com.co](mailto:juan.delahoz@relianz.com.co)

Juan Luis Perez  
CRC Engineer Assistant RELIZAN CAT  
Email: [practicante.desarrollocrc@relianz.com.co](mailto:practicante.desarrollocrc@relianz.com.co)

Reinaldo Peñaloza  
CRC Service Engineer RELIZAN CAT  
Email: [reinaldo.penalaza@relianz.com.co](mailto:reinaldo.penalaza@relianz.com.co)

Alejandro Cervantes  
CRC Intern RELIANZ CAT  
Email: [alejocerva@gmail.com](mailto:alejocerva@gmail.com)

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